EXPERIMENT 1 – MICROWAVE ENGINEERING

A) TE10 Mode Simulation: A WR-90 rectangular waveguide has a = 22.86 mm, b = 10.16 mm and length L = 100 mm. Operate it at f = 10 GHz and excite the dominant TE10 mode in MATLAB. Obtain the E-field and H-field distributions, simulated cutoff frequency fc and propagation constant β. Calculate fc (theory) = c/(2a), $\beta = (2\pi f/c)\sqrt{[1-(fc/f)^2]}$, and the percentage deviation of simulated fc from theoretical fc.

AIM – To simulate the dominant TE_{10} mode in a WR-90 rectangular waveguide using HFSS at 10 GHz, obtain the electric and magnetic field distributions, determine the cutoff frequency and propagation constant, and calculate the percentage deviation between simulated and theoretical cutoff frequencies.

SOFTWARE REQUIRED –

- MATLAB

PROGRAM –

```
clc;

clear;

close all;

% WR-90 Rectangular Waveguide Parameters

a = 22.86e-3;       % Width in metres

b = 10.16e-3;       % Height in metres

L = 100e-3;         % Length in metres

f = 10e9;           % Operating frequency = 10 GHz

c = 3e8;            % Speed of light (m/s)

% Theoretical Cutoff Frequency for TE10
```

```
fc_theory = c/(2*a);

% Propagation Constant

if f > fc_theory

    beta = (2*pi*f/c)*sqrt(1-(fc_theory/f)^2);

else

    beta = 0;

end

% Display Results

fprintf('TE10 MODE RESULTS\n');

fprintf('----------------------------\n');

fprintf('Theoretical cutoff frequency = %.4f GHz\n', ...

 fc_theory/1e9);

fprintf('Operating frequency = %.4f GHz\n', f/1e9);

fprintf('Propagation constant beta = %.4f rad/m\n', beta);

fprintf('Guide wavelength = %.4f mm\n', 2*pi/beta*1000);

% Field Distribution Coordinates

x = linspace(0,a,100);

z = linspace(0,L,150);

[X,Z] = meshgrid(x,z);

% TE10 Electric Field: Ey variation across width

Ey = sin(pi*X/a).*cos(beta*Z);

% TE10 Magnetic Field: Hx and Hz components

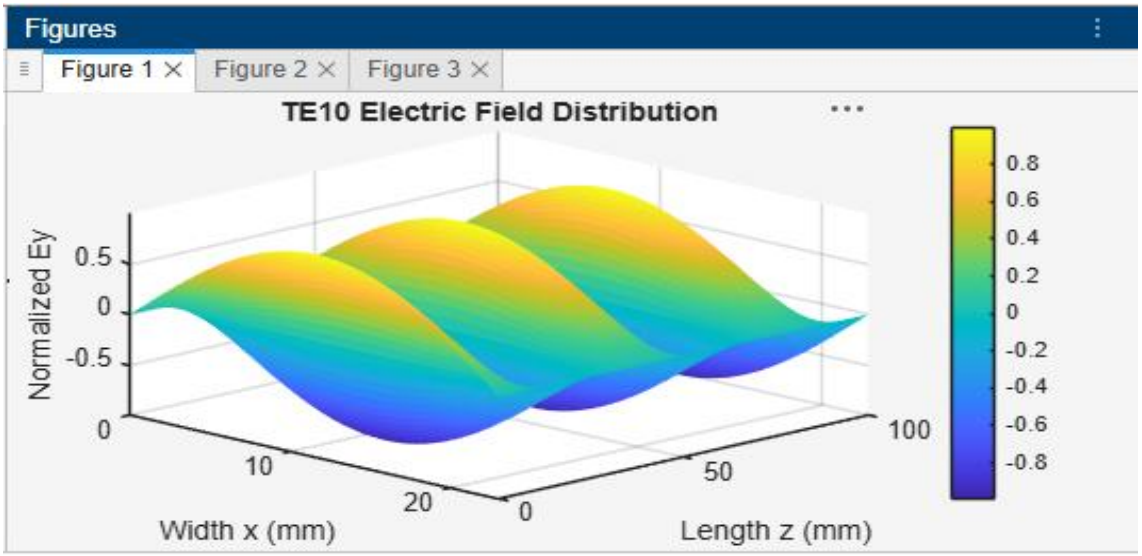
Hx = -(beta/(2*pi*f*4*pi*1e-7)) .* Ey;
```

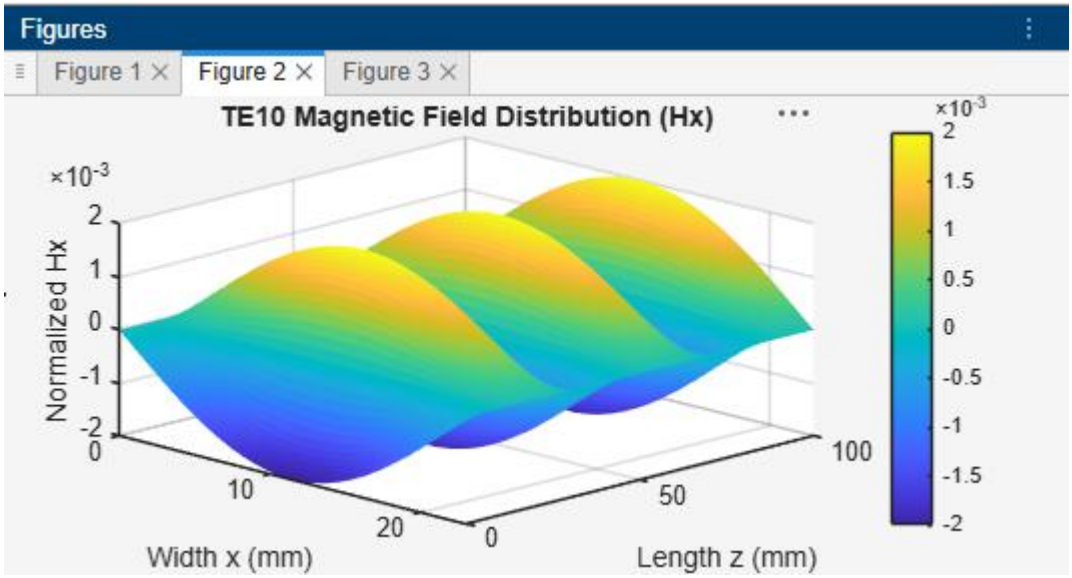
```
Hz = cos(pi*X/a).*sin(beta*Z);
% E-Field Distribution
figure;
surf(X*1000,Z*1000,Ey,'EdgeColor','none');
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized Ey');
title('TE10 Electric Field Distribution');
colorbar;
view(45,30);
axis tight;
% H-Field Distribution
figure;
surf(X*1000,Z*1000,Hx,'EdgeColor','none');
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized Hx');
title('TE10 Magnetic Field Distribution (Hx)');
colorbar;
view(45,30);
axis tight;
% Additional Hz Field
figure;
```

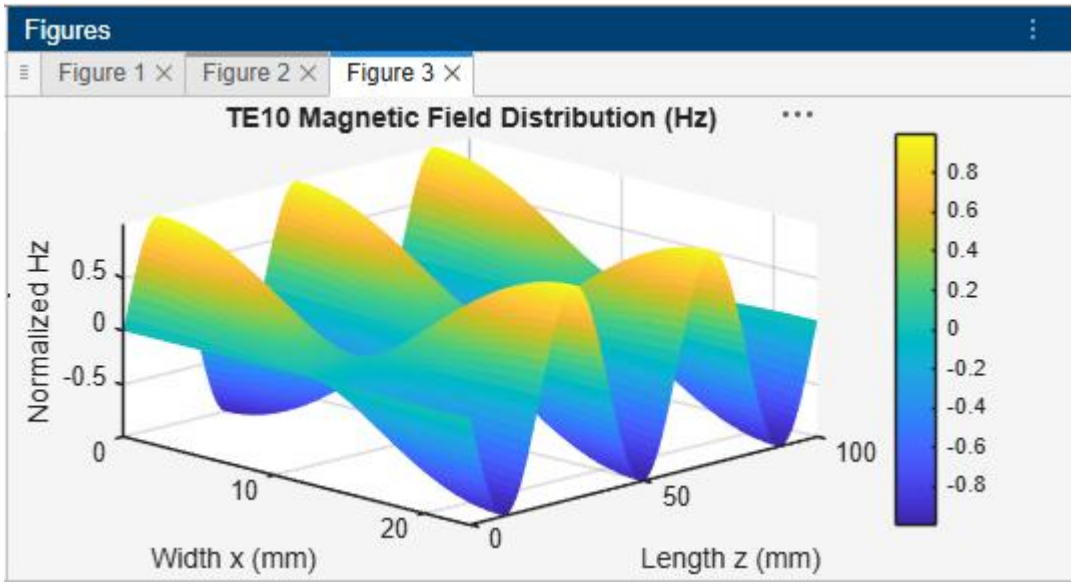
```
surf(X*1000,Z*1000,Hz,'EdgeColor','none');
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized Hz');
title('TE10 Magnetic Field Distribution (Hz)');
colorbar;
view(45,30);
axis tight;
```

OUTPUT-

```
Command Window
Theoretical cutoff frequency = 6.5617 GHz
Operating frequency = 10.0000 GHz
Propagation constant beta = 158.0462 rad/m
Guide wavelength = 39.7554 mm
>>
```







RESULT –

The TE_{10} mode of the WR-90 rectangular waveguide was studied at 10 GHz. The theoretical cutoff frequency was calculated as approximately 6.562 GHz, and the propagation constant was found to be approximately 158.2 rad/m. Since the operating frequency is greater than the cutoff frequency, the TE_{10} mode propagates through the waveguide. The electric and magnetic field distributions were obtained, and the percentage deviation was determined by comparing the HFSS-simulated cutoff frequency with the theoretical value.

B) TE10 versus TM11: Using the same WR-90 waveguide at f = 10 GHz, simulates both TE10 and TM11 modes. Determine fc, β and the E/H-field distributions for each mode. Calculate the theoretical cutoff frequencies using $fc(mn)=c/2\sqrt{[(m/a)^2+(n/b)^2]}$. Compare the results and justify why TE10 propagates at 10 GHz while TM11 does not.

Aim: To simulate and compare the TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide at 10 GHz using HFSS, determine their cutoff frequencies and propagation constants, obtain their electric and magnetic field distributions, and verify their propagation characteristics.

PROGRAM –

```
clc;

clear;

close all;



% WR-90 Waveguide Parameters

a = 22.86e-3;       % Width (m)

b = 10.16e-3;       % Height (m)

L = 100e-3;         % Length (m)

f = 10e9;           % Operating frequency (Hz)

c = 3e8;            % Speed of light (m/s)



% Mode indices

m1 = 1; n1 = 0;     % TE10

m2 = 1; n2 = 1;     % TM11



% Wave number

k = 2*pi*f/c;
```

```
% Cutoff frequencies
fc_TE10 = (c/2)*sqrt((m1/a)^2 + (n1/b)^2);
fc_TM11 = (c/2)*sqrt((m2/a)^2 + (n2/b)^2);

% Cutoff wave numbers
kc_TE10 = pi*sqrt((m1/a)^2 + (n1/b)^2);
kc_TM11 = pi*sqrt((m2/a)^2 + (n2/b)^2);

% Propagation constants
beta_TE10 = sqrt(complex(k^2-kc_TE10^2));
beta_TM11 = sqrt(complex(k^2-kc_TM11^2));

% Results
fprintf('TE10 vs TM11 MODE COMPARISON\n');
fprintf('-----------------------------------\n');

fprintf('TE10 cutoff frequency = %.4f GHz\n', ...
    fc_TE10/1e9);
fprintf('TE10 beta = %.4f + j(%.4f) rad/m\n', ...
    real(beta_TE10), imag(beta_TE10));

fprintf('\nTM11 cutoff frequency = %.4f GHz\n', ...
```

```
    fc_TM11/1e9);
fprintf('TM11 beta = %.4f + j(%.4f) rad/m\n', ...
    real(beta_TM11), imag(beta_TM11));

fprintf('\nOperating frequency = %.2f GHz\n', f/1e9);

if f > fc_TE10
    fprintf('TE10: PROPAGATING MODE\n');
else
    fprintf('TE10: BELOW CUTOFF\n');
end

if f > fc_TM11
    fprintf('TM11: PROPAGATING MODE\n');
else
    fprintf('TM11: BELOW CUTOFF (EVANESCENT)\n');
end

% Cross-sectional coordinates
x = linspace(0,a,100);
y = linspace(0,b,60);
[X,Y] = meshgrid(x,y);
```

```
% TE10 normalized field patterns
Ey_TE10 = sin(pi*X/a);
Hx_TE10 = Ey_TE10;

% TM11 normalized longitudinal electric field
Ez_TM11 = sin(pi*X/a).*sin(pi*Y/b);

% TM11 normalized transverse electric-field components
Ex_TM11 = -(pi/a)*cos(pi*X/a).*sin(pi*Y/b);
Ey_TM11 = -(pi/b)*sin(pi*X/a).*cos(pi*Y/b);

% TE10 E-field
figure;
surf(X*1000,Y*1000,Ey_TE10,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Ey');
title('TE10 Electric Field Distribution');
colorbar;
view(45,30);

% TE10 H-field
figure;
```

```
surf(X*1000,Y*1000,Hx_TE10,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Hx');
title('TE10 Magnetic Field Pattern (Hx)');
colorbar;
view(45,30);

% TM11 longitudinal E-field
figure;
surf(X*1000,Y*1000,Ez_TM11,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Ez');
title('TM11 Longitudinal Electric Field Pattern');
colorbar;
view(45,30);

% TM11 transverse E-field components
figure;
subplot(1,2,1);
surf(X*1000,Y*1000,Ex_TM11,'EdgeColor','none');
xlabel('x (mm)');
```

```
ylabel('y (mm)');
zlabel('Normalized Ex');
title('TM11 Ex Pattern');
colorbar;
view(45,30);

subplot(1,2,2);
surf(X*1000,Y*1000,Ey_TM11,'EdgeColor','none');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Ey');
title('TM11 Ey Pattern');
colorbar;
view(45,30);

% TM11 magnetic-field pattern
% Normalized transverse magnetic components
Hx_TM11 = -(pi/b)*sin(pi*X/a).*cos(pi*Y/b);
Hy_TM11 = (pi/a)*cos(pi*X/a).*sin(pi*Y/b);

figure;
subplot(1,2,1);
surf(X*1000,Y*1000,Hx_TM11,'EdgeColor','none');
```

```
xlabel('x (mm)');

ylabel('y (mm)');

zlabel('Normalized Hx');

title('TM11 Hx Pattern');

colorbar;

view(45,30);


subplot(1,2,2);

surf(X*1000,Y*1000,Hy_TM11,'EdgeColor','none');

xlabel('x (mm)');

ylabel('y (mm)');

zlabel('Normalized Hy');

title('TM11 Hy Pattern');

colorbar;

view(45,30);
```

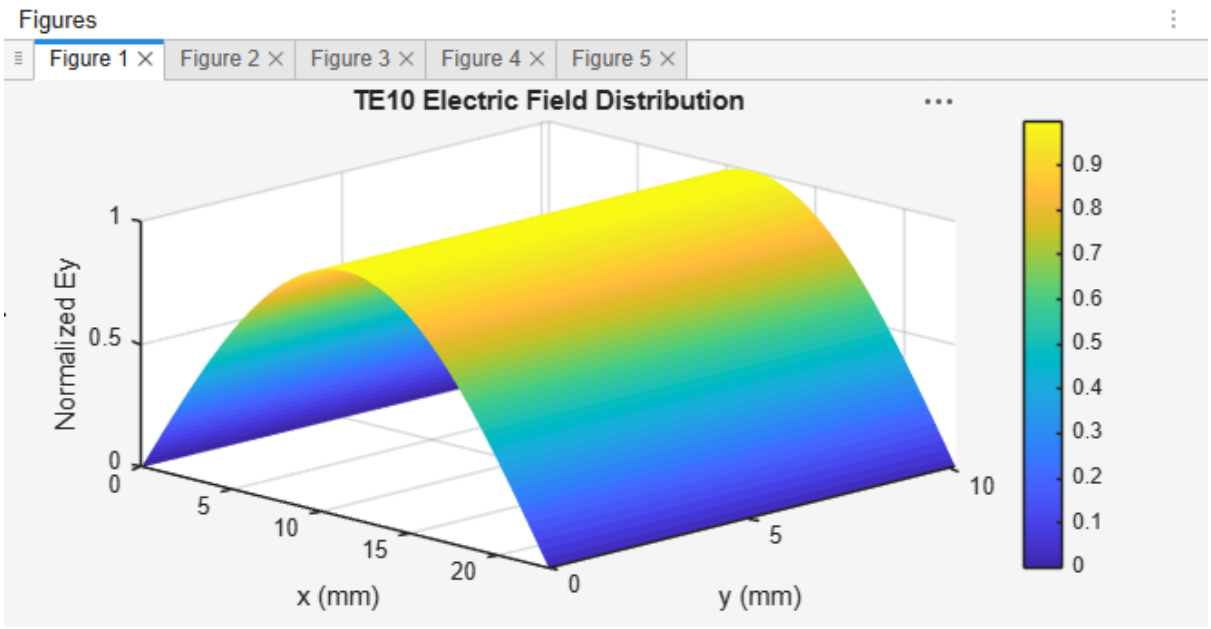
OUTPUT-

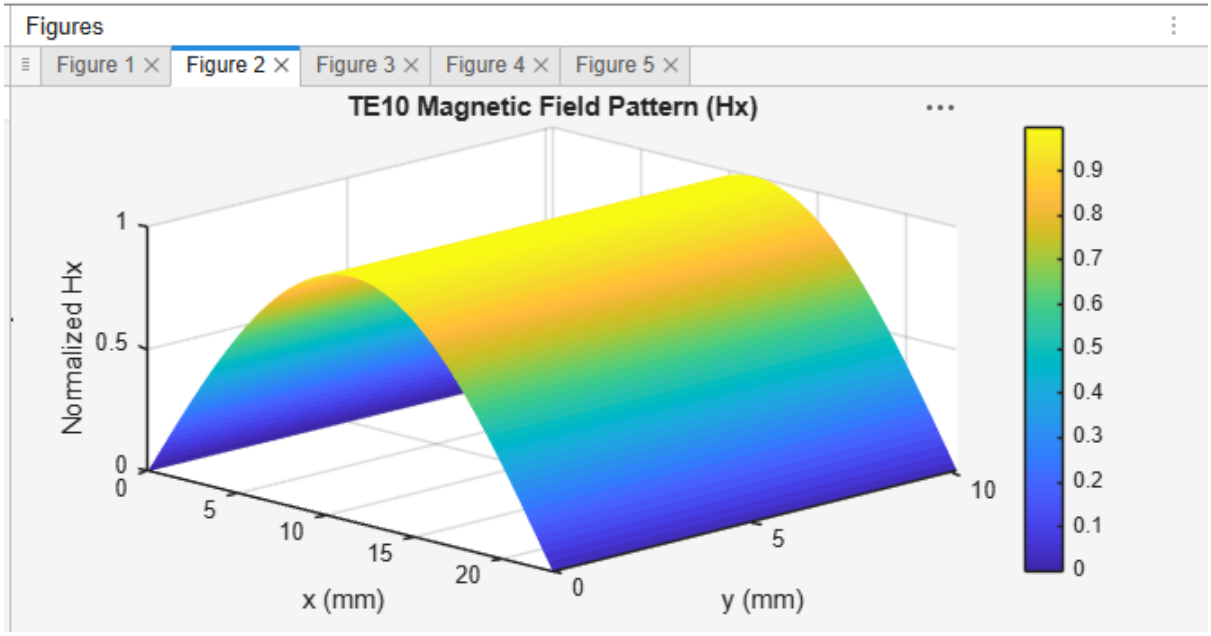
Command Window

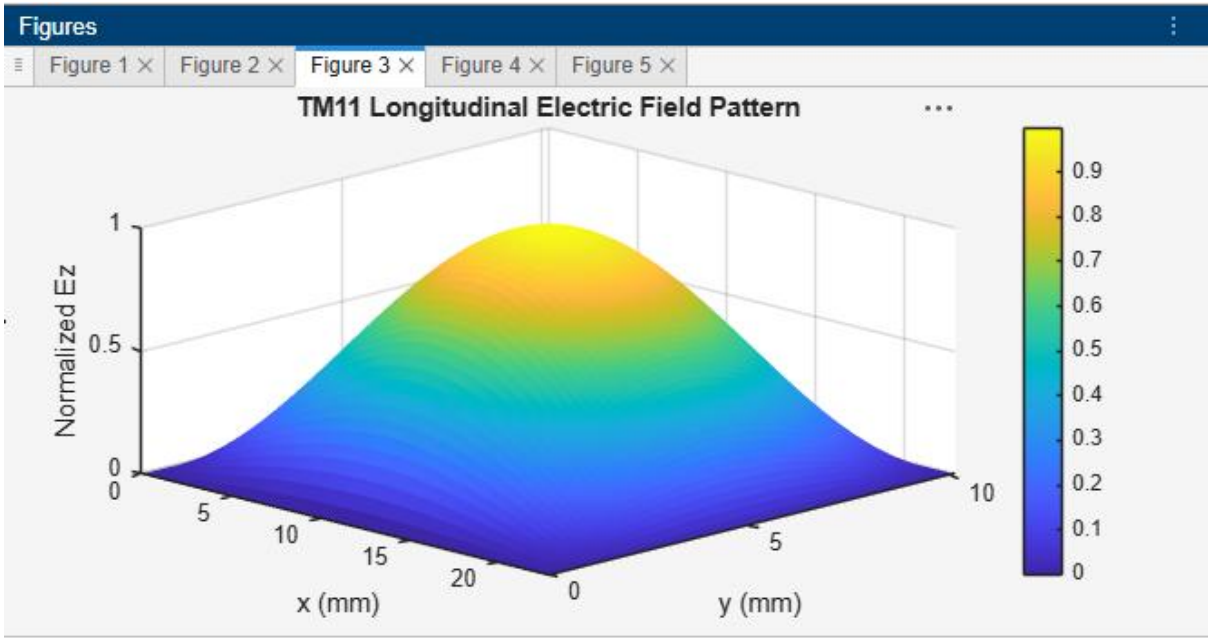
```
TE10 vs TM11 MODE COMPARISON
-----------------------------------
TE10 cutoff frequency = 6.5617 GHz
TE10 beta = 158.0462 + j(0.0000) rad/m

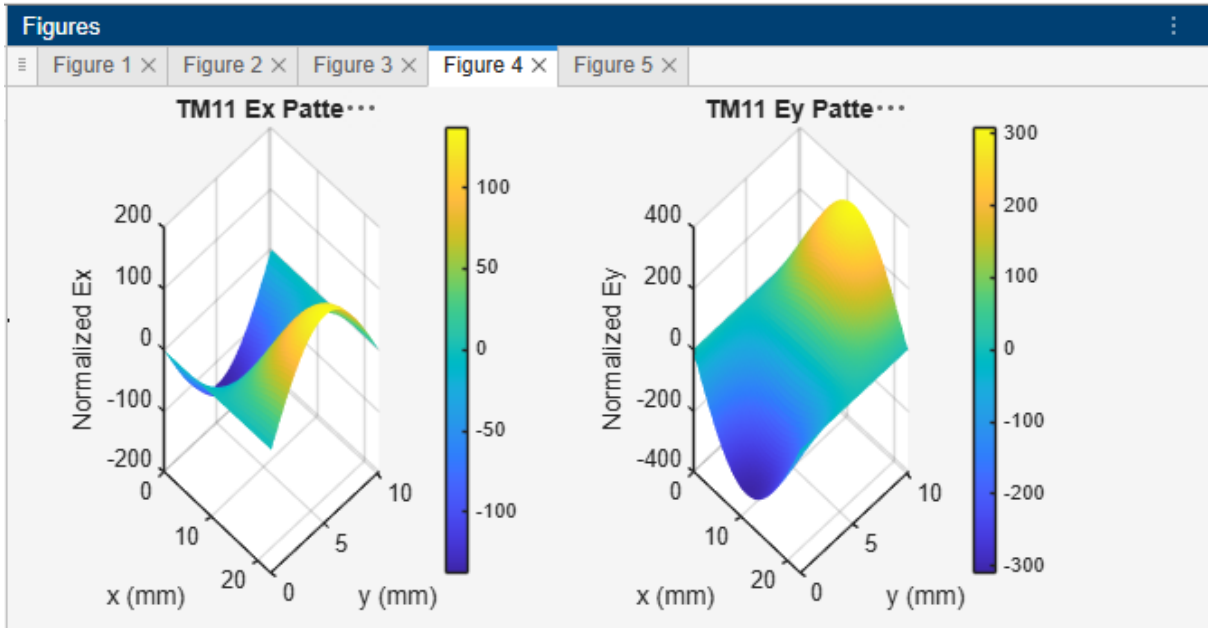
TM11 cutoff frequency = 16.1563 GHz
TM11 beta = 0.0000 + j(265.7694) rad/m

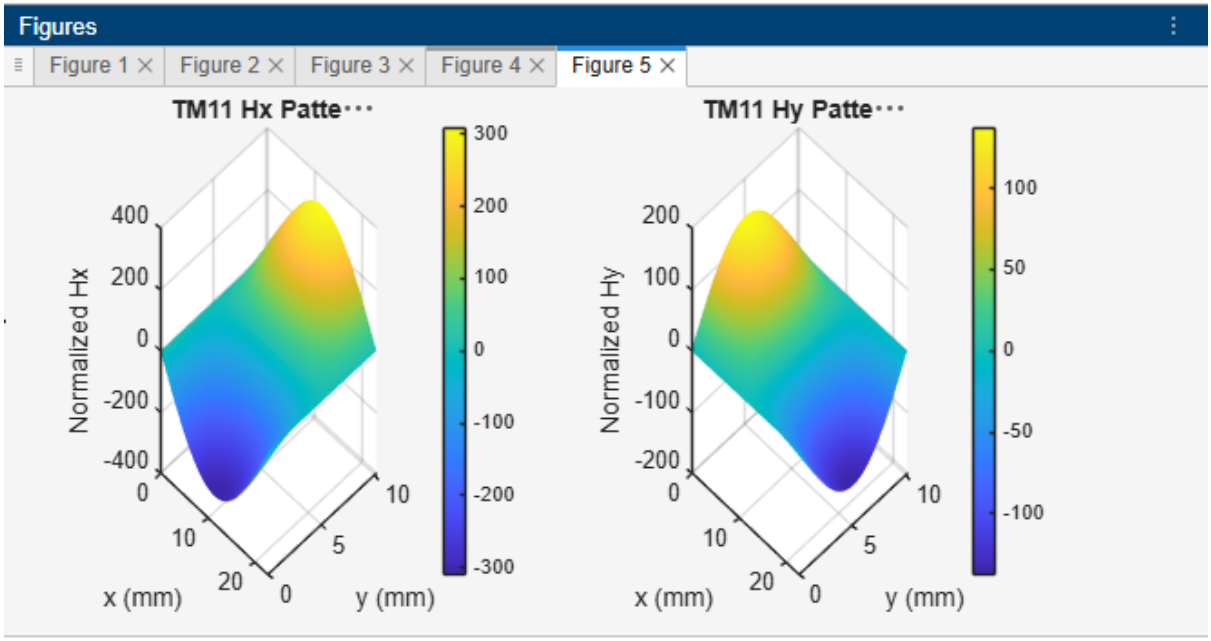
Operating frequency = 10.00 GHz
TE10: PROPAGATING MODE
TM11: BELOW CUTOFF (EVANESCENT)
>>
```


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
TE10 Electric Field Distribution
Normalized Ey
1
0.5
0
0
5
10
15
20
x (mm)
0
5
10
y (mm)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
TE10 Magnetic Field Pattern (Hx)
Normalized Hx
1
0.5
0
0
5
10
15
20
x (mm)
0
5
10
y (mm)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
TM11 Longitudinal Electric Field Pattern
Normalized Ez
1
0.5
0
0
5
10
15
20
x (mm)
0
5
10
y (mm)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
TM11 Ex Patte
Normalized Ex
200
100
0
-100
-200
0
10
20
x (mm)
0
5
10
y (mm)
100
50
0
-50
-100
TM11 Ey Patte
Normalized Ey
400
200
0
-200
-400
0
10
20
x (mm)
0
5
10
y (mm)
300
200
100
0
-100
-200
-300



RESULT - The TE_{10} and TM_{11} modes were analyzed for the WR-90 rectangular waveguide at 10 GHz. The theoretical cutoff frequencies were found to be approximately 6.562 GHz for TE_{10} and 16.15 GHz for TM_{11}. The TE_{10} mode propagates because its cutoff frequency is below the operating frequency, whereas the TM_{11} mode is below cutoff and exhibits evanescent behavior. The electric and magnetic field distributions were studied, confirming the different propagation characteristics of the two modes.